

Material Safety Data Sheet

1. PRODUCT AND COMPANY IDENTIFICATION

Product name Tris(hydroxymethyl)aminomethane
Supplier : Longlife Biochemical Co., Ltd.
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2. HAZARDS IDENTIFICATION

Classification of the substance or mixture

Not a hazardous substance or mixture according to Regulation (EC) No 1272/2008

This substance is not classified as dangerous according to Directive 67/548/EEC.

Label elements

The product does not need to be labelled in accordance with EC directives or respective national laws.

Other hazards

This substance is not considered to be persistent, bioaccumulating nor toxic (PBT).

3. COMPOSITION/INFORMATION ON INGREDIENTS

Synonyms: Tris base, Tromethamine
Formula: $C_4H_{11}NO_3$
Molecular Weight: 121.14g/mol

Component	Concentration
Tris(hydroxymethyl)aminomethane	
CAS-No.	77-86-1
EC-No.	201-064-4

No components need to be disclosed according to the applicable regulations.

4. FIRST AID MEASURES

Description of first aid measures

If inhaled If breathed in, move person into fresh air. If not breathing, give artificial respiration.

In case of skin contact Wash off with soap and plenty of water.

In case of eye contact Flush eyes with water as a precaution.

If swallowed Never give anything by mouth to an unconscious person. Rinse mouth with water.

Most important symptoms and effects, both acute and delayed

The most important known symptoms and effects are described in the labelling (see section 2.2) and/or in section 11

Indication of any immediate medical attention and special treatment needed

no data available

5. FIREFIGHTING MEASURES

Suitable extinguishing media	Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.
Special hazards arising from the substance or mixture	no data available
Advice for firefighters	Wear self contained breathing apparatus for fire fighting if necessary.
Further information	No data available.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions	Avoid dust formation. Avoid breathing vapours, mist or gas. For personal protection see section 8.
Environmental precautions	Do not let product enter drains.
Methods and materials for containment and cleaning up	Sweep up and shovel. Keep in suitable, closed containers for disposal.
Reference to other sections	For disposal see section 13.

7. HANDLING AND STORAGE

Precautions for safe handling	Provide appropriate exhaust ventilation at places where dust is formed. For precautions see section 2.2.
Conditions for safe storage	Store in cool place. Keep container tightly closed in a dry and well-ventilated place.
Specific end use(s)	A part from the uses mentioned in section 1.2 no other specific uses are stipulated

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters

Components with workplace control parameters

Exposure controls

Appropriate engineering controls

General industrial hygiene practice.

Personal protective equipment

Eye/face protection

Use equipment for eye protection tested and approved under appropriate government standards

such as NIOSH (US) or EN 166(EU).

Skin protection

Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique

(without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices.

Wash and dry hands.

The selected protective gloves have to satisfy the specifications of EU Directive 89/686/EEC and

the standard EN 374 derived from it.

Full contact

Material: Nitrile rubber

Minimum layer thickness: 0.11 mm

Break through time: > 480 min

Splash protection

Material: Nitrile rubber

Minimum layer thickness: 0.11 mm

Break through time: > 30 min

Body protection

Choose body protection in relation to its type, to the concentration and amount of dangerous substances, and to the specific work-place., The type of protective equipment must be selected

according to the concentration and amount of the dangerous substance at the specific workplace.

Respiratory protection

Respiratory protection is not required. Where protection from nuisance levels of dusts are desired, use type N95 (US) or type P1 (EN 143) dust masks. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

Control of environmental exposure

Do not let product enter drains.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance	Form: crystalline Colour: colourlesswhite
Odour	odourless
Odour Threshold	no data available
pH (in 1% solution)	10.0-11.5
Melting point/freezing point	Melting point/range: 168 °C~171°C
Initial boiling point and boiling range	288 °C at 1.013 hPa - Decomposes below the boiling point.
Flash point	no data available
Evaporation rate	no data available
Flammability (solid, gas)	no data available
Upper/lower flammability or explosive limits	no data available
Vapour pressure	no data available
Vapour density	no data available
Relative density	no data available
Water solubility	678 g/l at 20 °C
Partition coefficient: octanol/water	log Pow: -2,31 at 20 °C

Auto-ignition temperature	no data available
Decomposition temperature	no data available
Viscosity	no data available
Explosive properties	no data available
Oxidizing properties	no data available
Other safety information	no data available
Bulk density	800kg/m ³
Dissociation constant	8,22 at 25 °C

10. STABILITY AND REACTIVITY

Reactivity	no data available
Chemical stability	Stable under recommended storage conditions.
Possibility of hazardous reactions	no data available
Conditions to avoid	hygroscopic
Incompatible materials	Strong oxidizing agents
Hazardous decomposition products	Other decomposition products - no data available - In the event of fire: see section 5

11. TOXICOLOGICAL INFORMATION

Acute toxicity	
LD50	LD50 Oral - rat - > 3.000 mg/kg LD50 Dermal - rat - > 5.000 mg/kg (OECD Test Guideline 402)
Skin corrosion/irritation	Skin - rabbit Result: No skin irritation (OECD Test Guideline 404)
Serious eye damage/eye irritation	Eyes - rabbit Result: No eye irritation (OECD Test Guideline 405)
Respiratory or skin sensitization	Buehler Test - guinea pig Does not cause skin sensitisation. (OECD Test Guideline 406)
Germ cell mutagenicity	Result: Not mutagenic in Ames Test. in vitro assay Result: negative In vitro tests did not show mutagenic effects Result: In vivo tests did not show any chromosomal changes.
Carcinogenicity	
IARC: No component of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.	
Reproductive toxicity	no data available

Specific target organ toxicity - single exposure	no data available
Specific target organ toxicity - repeated exposure	no data available
Aspiration hazard	no data available
Additional Information	Repeated dose toxicity - rat - Oral - No observed adverse effect level - 1.000 mg/kg RTECS: TY2900000

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

12. ECOLOGICAL INFORMATION

Toxicity	
Toxicity to daphnia and other aquatic invertebrates	EC50 - Daphnia - > 980 mg/l - 48 h
Toxicity to algae	EC50 - Algae - 397 mg/l - 72 h NOEC - Algae - 100 mg/l - 72 h
Persistence and degradability	
Biodegradability	Result: - Readily biodegradable. (OECD Test Guideline 301F)
Bioaccumulative potential	No bioaccumulation is to be expected (log Pow ≤ 4).
Mobility in soil	no data available
Results of PBT and vPvB assessment	This substance is not considered to be persistent, bioaccumulating nor toxic (PBT).
Other adverse effects	no data available

13. DISPOSAL CONSIDERATIONS

Waste treatment methods	
Product	Offer surplus and non-recyclable solutions to a licensed disposal company.
Contaminated packaging	Dispose of as unused product.

14. TRANSPORT INFORMATION

DOT (US)	Not dangerous goods
IMDG	Not dangerous goods
IATA	Not dangerous goods

15. REGULATORY INFORMATION

This safety datasheet complies with the requirements of Regulation (EC) No. 1907/2006.

Safety, health and environmental regulations/legislation specific for the substance or mixture

no data available

Chemical Safety Assessment

For this product a chemical safety assessment was not carried out

16. OTHER INFORMATION

Further information

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